MENOPAUSE PROXY + SNS/LIPOLYSIS IN HUMAN ADIPOSE (GTEX V10)

2026-02-02

Goal + constraints

- Goal: add *human adipose* transcriptomic evidence consistent with adrenergic/lipolysis differences in postmenopause.
- Constraints:
  - GTEx is baseline/unstimulated tissue → acute SNS transcripts may be weak.
  - Chronic SNS can yield catecholamine resistance → directionality not straightforward.
  - Bulk adipose is mixed cell types → composition can dominate signals.
- Definition here: ‘SNS activation’ = increased sympathetic/adrenergic drive to adipose; in RNA this is often proxied by immediate-early genes (NR4A/FOS/JUN/EGR1) and downstream cAMP/lipolysis programs, but baseline tissue may not capture acute responses.

Data + cohorts (what “pre / peri / post” mean here)

- Dataset: **GTEx v10** bulk adipose RNA-seq (counts), analyzed separately by depot.
- Women-only samples:
  - Subcutaneous: n=233
  - Visceral (omentum): n=180
- Menopause proxy by **age-bin midpoint** (GTEx provides age bins, not true menopause status):
  - **pre**: age_mid < 45
  - **peri**: 45 ≤ age_mid ≤ 55
  - **post**: age_mid > 55
- Note (important for interpretation): GTEx age is **10-year bins**; with this midpoint-based rule, groups map to **pre=20–49**, **peri=50–59**, **post=60+**.
- “Strict pre vs post” analyses exclude peri (50–59) to focus on likely extremes.
  - ? Is this acceptable?

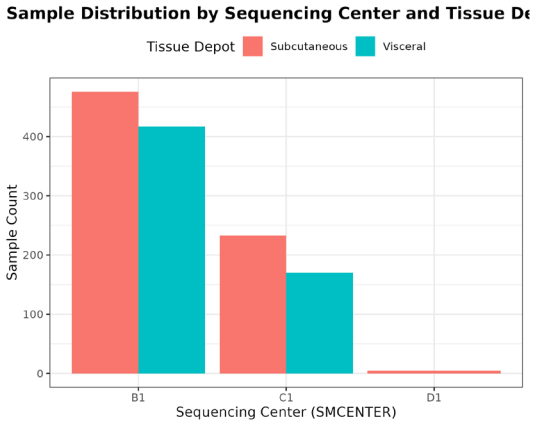
GTEx “batch” reality: sequencing center and batches are major covariates

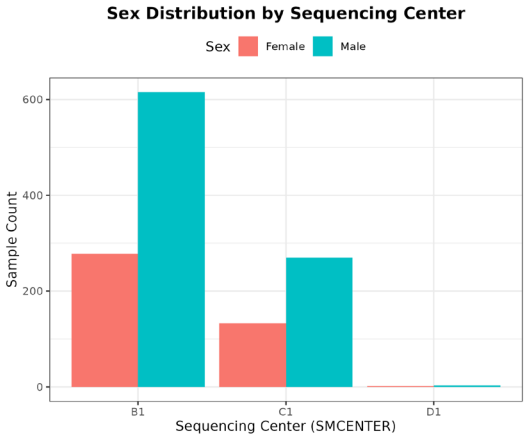
In GTEx adipose, technical variables are not independent of biology:

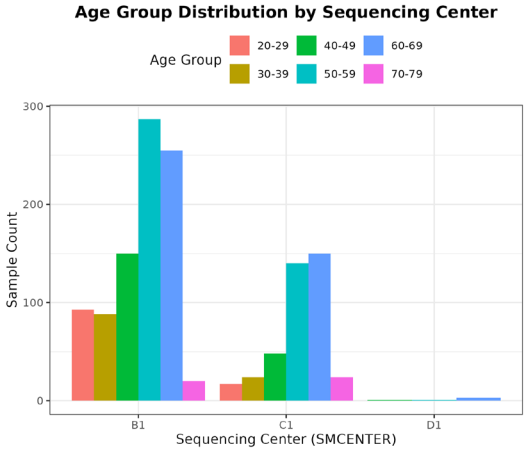
- **SMCENTER** (sequencing center) differs by **depot**, **sex**, and **age-bin distribution**.
- “Batch correction” is not a single plug-and-play step because aggressive removal can also remove biology if batch is confounded with group.

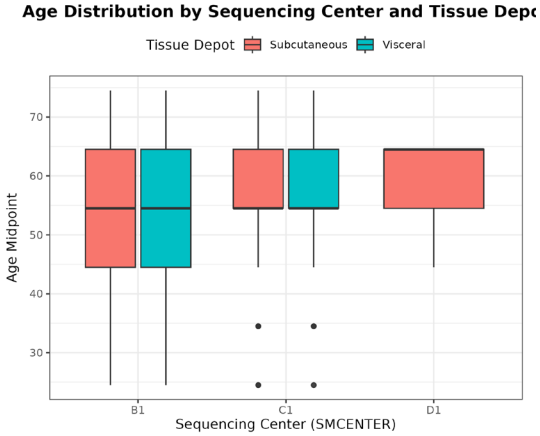
Therefore in downstream models we:

- Always include **SMCENTER** as a covariate.
- Treat “gene-expression batch” (e.g., **SMGEBTCH**) as a **sensitivity/robustness** adjustment (not a single “best” model).





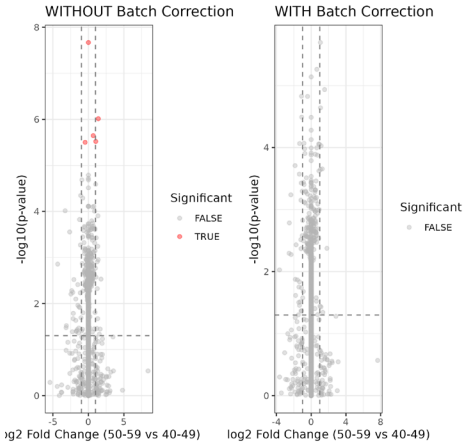


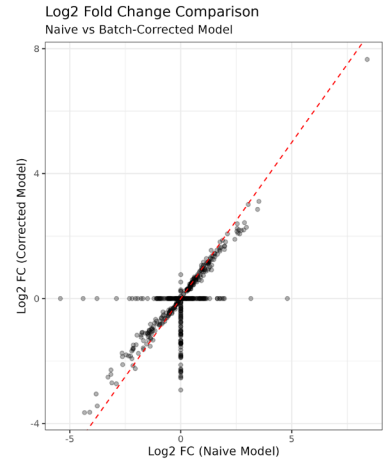


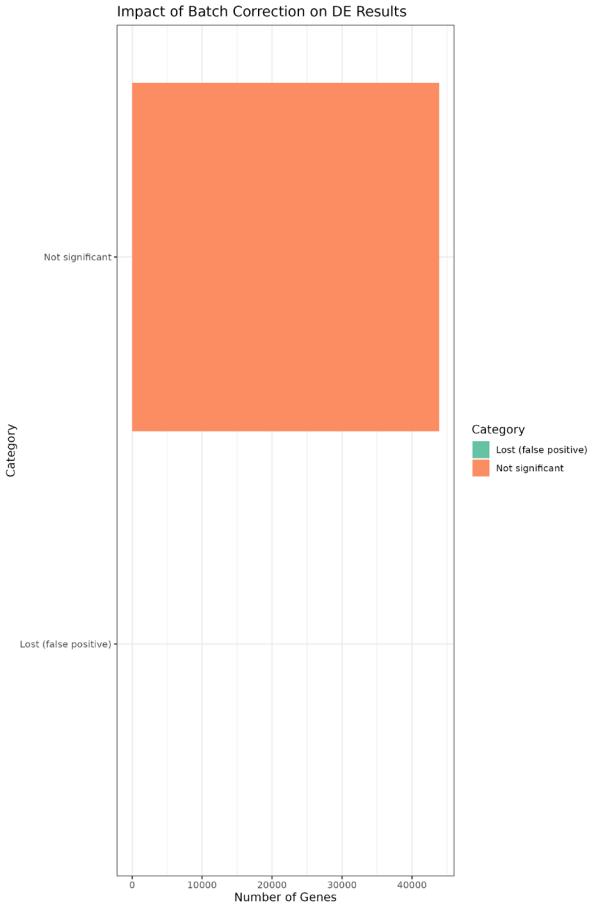
Why we don’t just “ComBat everything”: batch adjustment changes conclusions

Example (female subcutaneous; 40–49 vs 50–59): naive DE can yield “significant” genes that disappear after adding **SMCENTER** (likely false positives).

Visualization: left shows naive vs center-adjusted DE comparison; right shows PCA before/after ComBat (illustrates how strong center-related structure is).







How gene-set “z-scores” were computed (for Slides 4–6)

For each depot (women-only), using VST-normalized expression:

Step 1: DESeq2 VST on counts.

Step 2: map Ensembl → gene symbols; collapse duplicates by mean.

Step 3: for each gene in a gene set, compute per-sample z-score across samples: - z(gene, sample) = (expr − mean_over_samples) / sd_over_samples

Step 4: gene-set score per sample = mean of z(gene, sample) across genes in the set.

Interpretation: a higher score means **higher relative expression of the genes in that set** *within that depot’s cohort* (not an absolute activity measurement).

- Kenichi panel gene sets (exact gene lists): references/kenichi_panel_gene_sets.tsv
- Plot convention: each dot = one GTEx sample’s gene-set score (not one gene).

Kenichi panel — subcutaneous baseline gene-set activity

Conclusion (SMCENTER-adjusted):

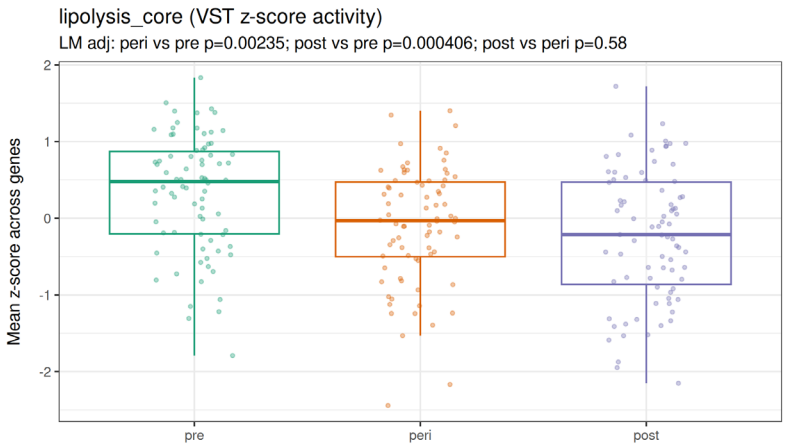
Groups shown: pre=20–49, peri=50–59, post=60+ (age-bin proxy; not true menopause status).

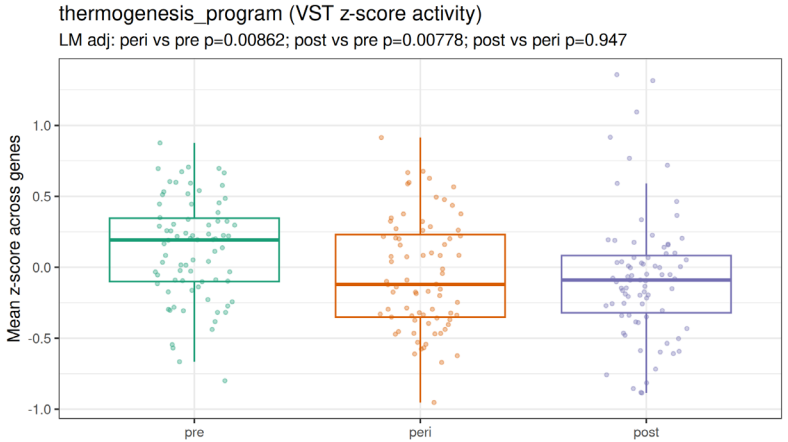
Score shown: mean gene-wise z-score across the genes in each set (VST expression; z computed across women in this depot).

Kenichi panel includes multiple gene sets; this figure shows lipolysis_core and thermogenesis_program (program-level sets).

Each dot = one sample; per-sample score = mean(z) across the genes in the set.

Acute SNS immediate-early set: not robust at baseline





Kenichi panel — visceral baseline gene-set activity

Conclusion (SMCENTER-adjusted):

Groups shown: pre=20–49, peri=50–59, post=60+ (age-bin proxy; not true menopause status).

Score shown: mean gene-wise z-score across the genes in each set (VST expression; z computed across women in this depot).

Kenichi panel includes multiple gene sets; this figure shows lipolysis_core and thermogenesis_program (program-level sets).

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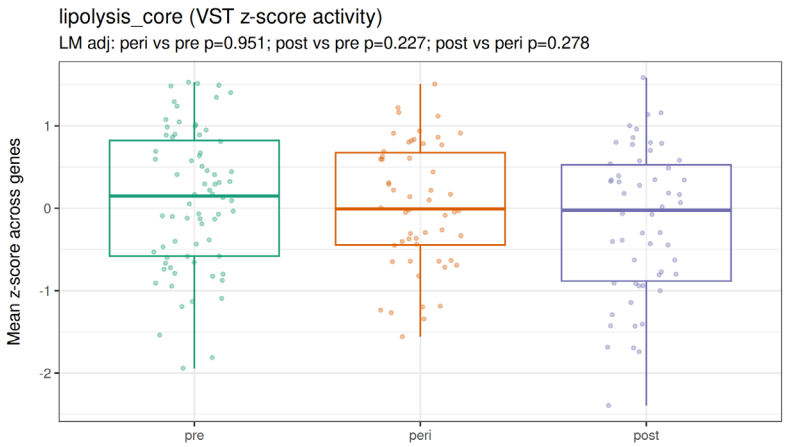
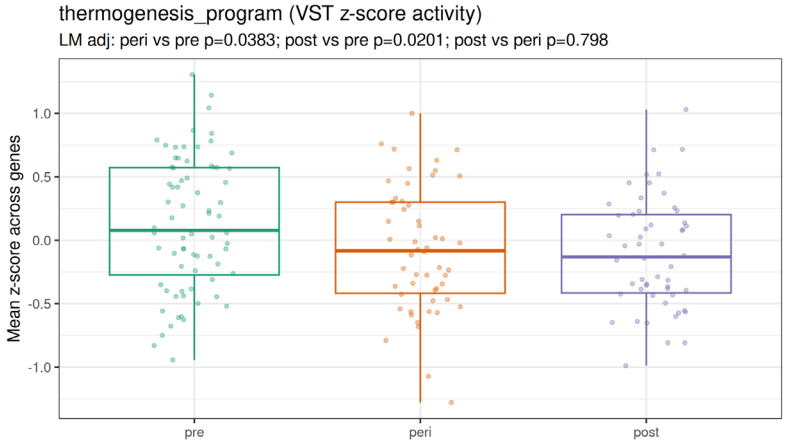
Acute SNS immediate-early set: not robust at baseline



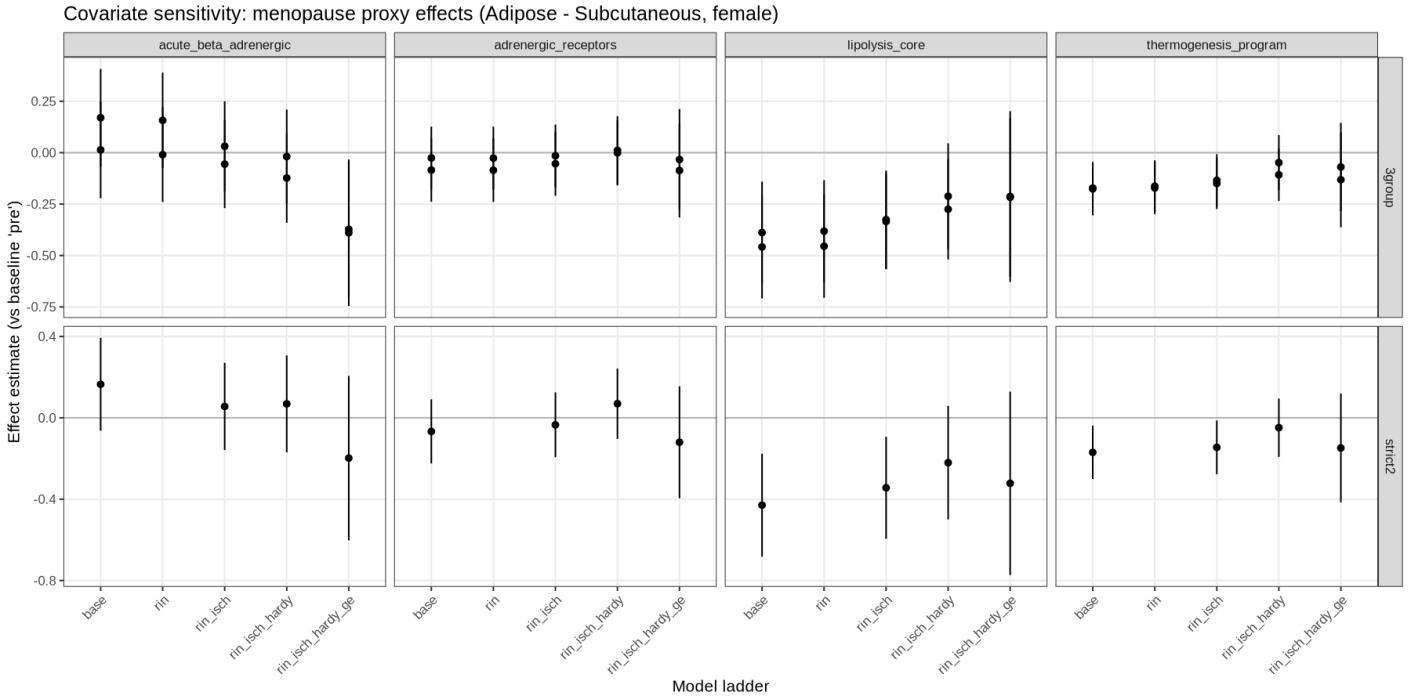


Confounding sensitivity analyses (what Slides 10–12 show)

- These slides are NOT classical GSEA outputs.
- We compute per-sample gene-set/module scores from VST expression (mean gene-wise z across set genes).
- Then we fit linear models for group contrasts under a covariate ‘ladder’:
- - base: SMCENTER + group
- - add technical covariates: SMRIN (RIN), SMTSISCH (ischemic time), DTHHRDY, and optionally SMGEBTCH (gene-expression batch)
- - add composition proxies: adipocyte / macrophage / fibroblast marker scores
- Interpretation: if an effect disappears after adding composition proxies, the bulk-tissue signal is likely driven by cell-type proportion shifts (not necessarily per-adipocyte transcriptional change).

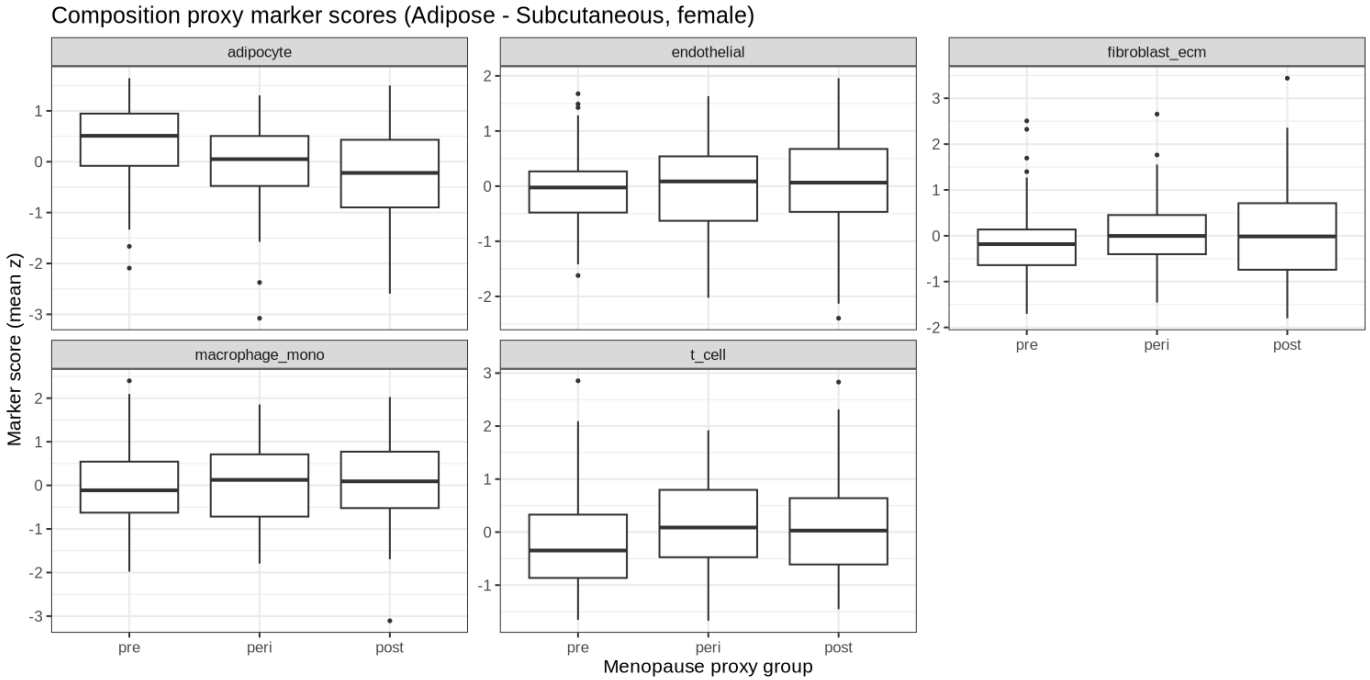
Technical confounding sensitivity (RIN / ischemic time / Hardy / batch)

Purpose: test whether menopause-proxy effects are stable when adding GTEx technical covariates.



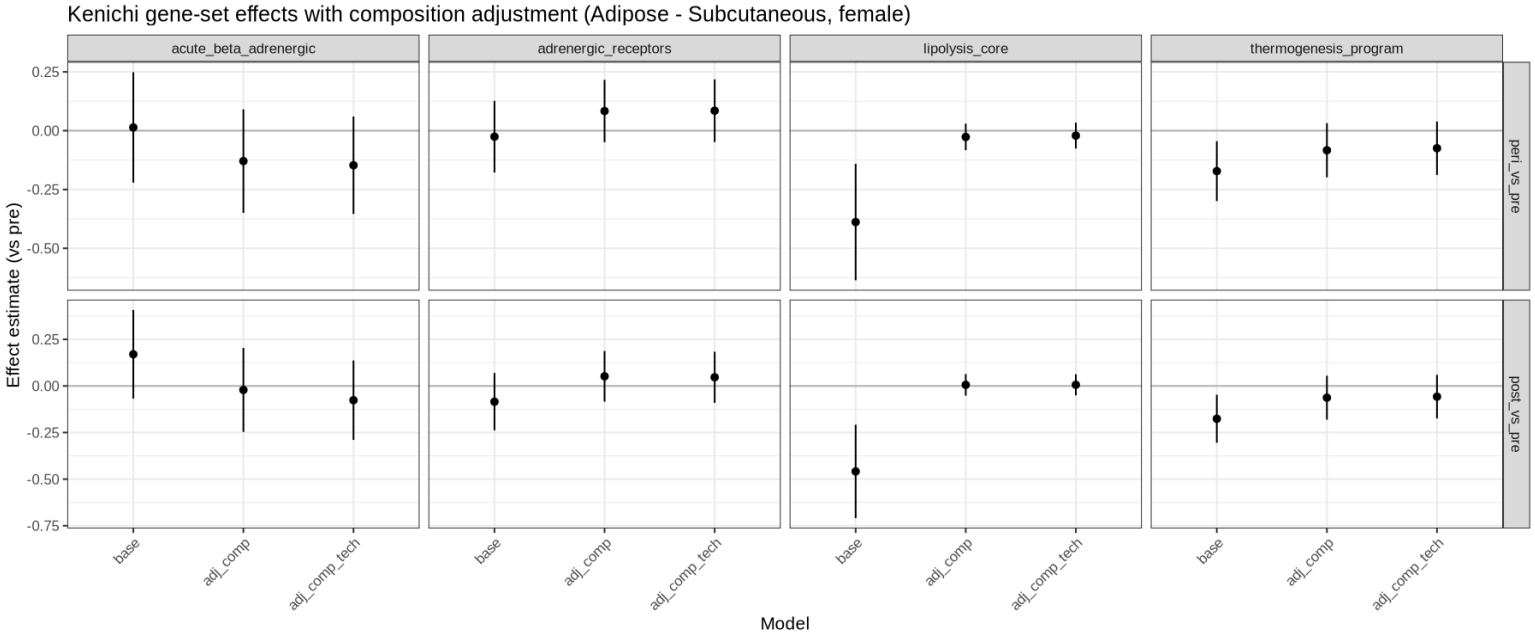
Bulk adipose composition proxies shift with menopause proxy

Key point: adipocyte-content proxy decreases in peri/post vs pre in subcutaneous bulk tissue.



Subcutaneous: composition adjustment explains “lipolysis_core down”

Key point: the subcutaneous lipolysis_core “down” signal is strongly composition-sensitive.



Quick summary (confounding + composition): interpretation

- Bulk adipose RNA is a mixture: bulk_expr ≈ p_adip·expr_adip + p_SVF·expr_SVF.
- So if adipocyte fraction (p_adip) decreases, adipocyte-enriched genes/sets can look ‘down’ in bulk even when per-adipocyte expression is unchanged.
- In subcutaneous GTEx, adipocyte proxy score decreases in peri/post vs pre.
- After adding composition proxies as covariates, the residual post–pre difference for lipolysis_core is ~0 → we cannot claim strong per-adipocyte downregulation from bulk alone.
- This does NOT prove there is no per-adipocyte dysregulation; it means bulk data cannot separate composition vs per-cell regulation without additional evidence (e.g., purified adipocytes or single-cell / deconvolution).

Subcutaneous: key table (post vs pre) — base vs composition-adjusted

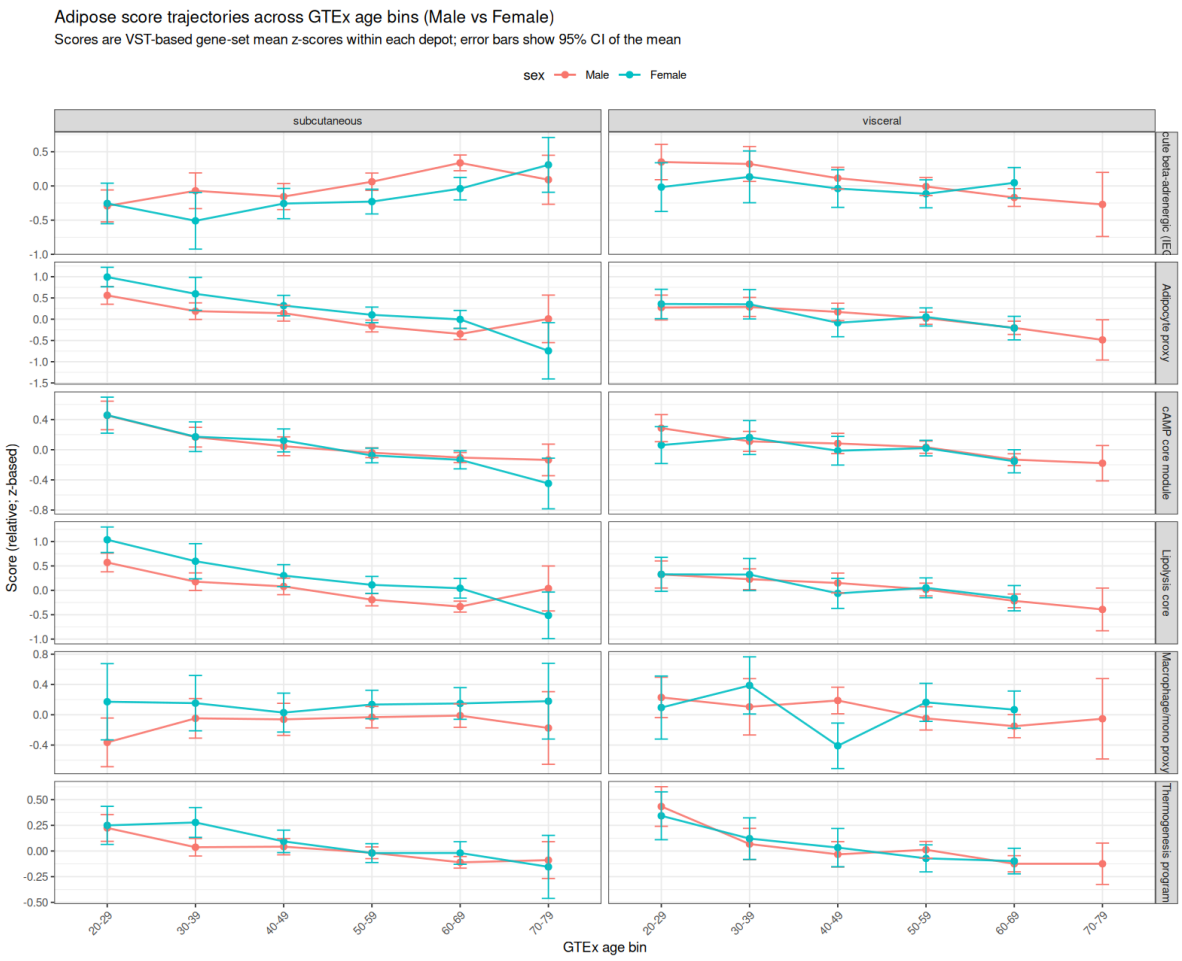
Subcutaneous: Kenichi gene-set effect (post vs pre)
Base vs composition-adjusted models

	Base (SMCENTER)	+Composition	+Comp+Tech
acute_beta_adrenergic	0.170 (p=0.162)	-0.021 (p=0.852)	-0.077 (p=0.482)
adrenergic_receptors	-0.085 (p=0.283)	0.052 (p=0.457)	0.047 (p=0.506)
lipolysis_core	-0.459 (p=4.1e-04)	0.006 (p=0.845)	0.006 (p=0.830)
thermogenesis_program	-0.176 (p=0.008)	-0.063 (p=0.296)	-0.058 (p=0.336)

Male vs female context (Kenichi caveat) — trajectories across age bins

Kenichi note: male–female comparisons are not a perfect control (men also have hormonal aging + psychosocial confounds), but can provide context.

Here we plot key adipose scores across GTEx age bins in **both sexes** (separate depots).



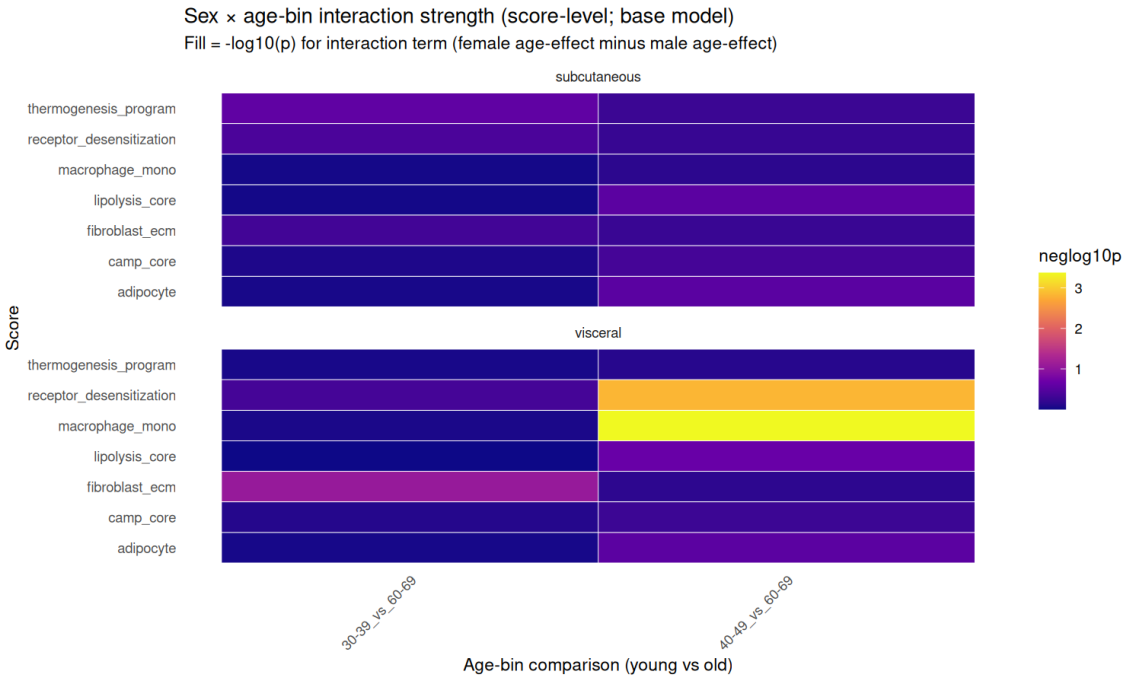
Male vs female context — sex×age interaction (score-level)

Model: score ~ SMCENTER + sex + age_group + sex:age_group (base); comparisons: 40–49 vs 60–69 and 30–39 vs 60–69.

Color = −log10(p) for interaction term (female aging effect minus male aging effect).

Interaction term = (Female_old − Female_young) − (Male_old − Male_young) = ‘female age-effect minus male age-effect’ (difference-in-differences).

Summary: several aging effects appear stronger in males (consistent with the “male aging may be larger” caveat); female-specific interaction is modest for most scores after composition-aware adjustment.



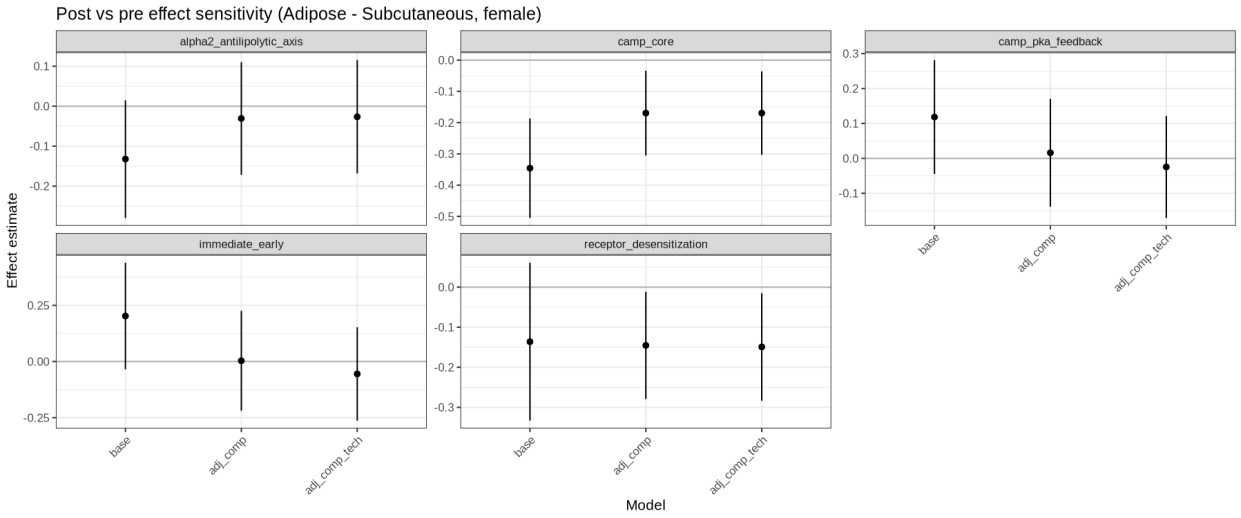
Adrenergic/cAMP/desensitization modules (subcutaneous)

Interpretation: more compatible with altered responsiveness/chronic adaptation than acute SNS activation at baseline.

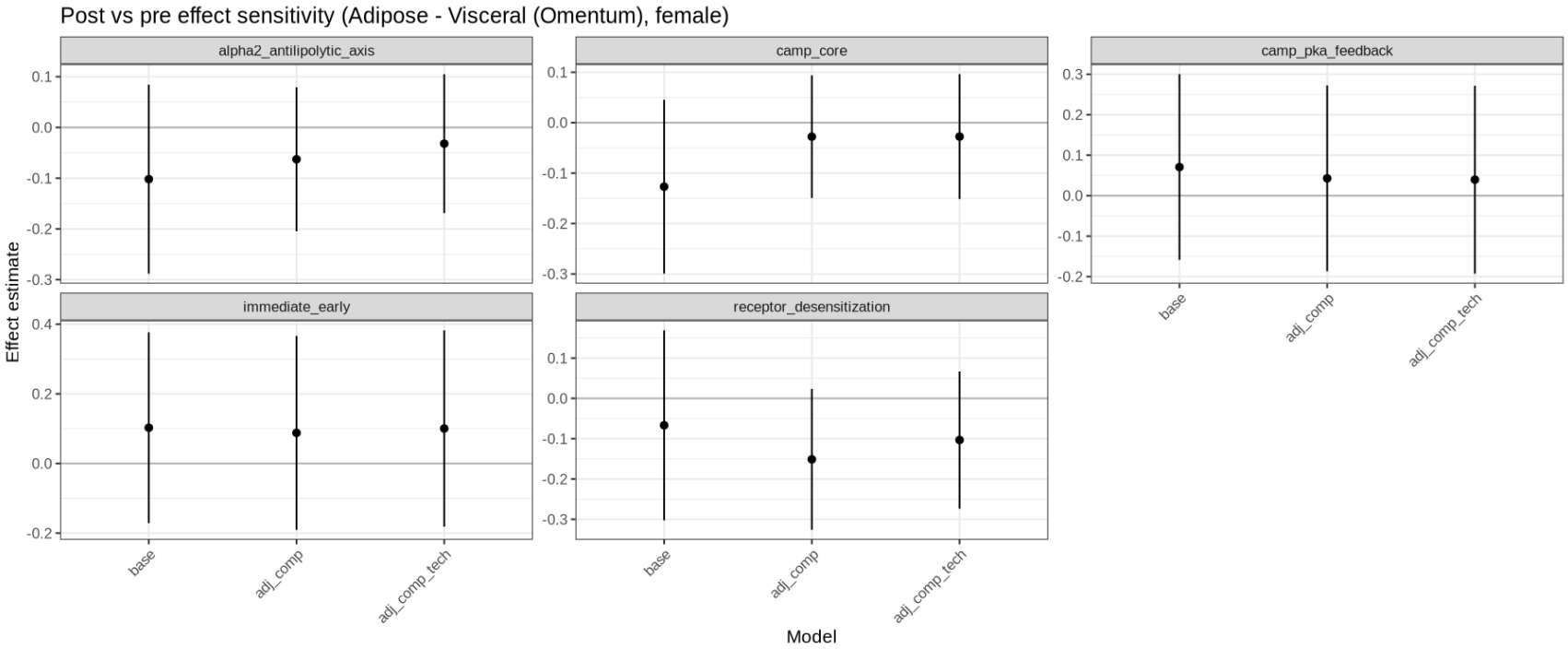
Effect estimate = adjusted difference in module score (post − pre) from linear models.

Models shown: base = SMCENTER + group; adj_comp = base + (adipocyte/macrophage/fibroblast proxies); adj_comp_tech = adj_comp + (SMRIN + SMTSISCH).

Positive = higher in post; negative = lower in post (relative expression of genes in the module).



Adrenergic/cAMP/desensitization modules (visceral)



Menopause signature: senMayoLike v2 (curated, evidence-backed)

Signature size: 13 total genes (12 up in post, 1 down in post).

How selected (summary):

- Start: GTEx subcutaneous menopause-proxy discovery list (Approach 1; 81 genes).

- Remove lncRNA/pseudogene/keratin/blood-heavy markers; score robustness to composition + technical covariates.

- v1 = data-driven ‘robustness-screen’ subset (top genes in adj_comp_tech model).

- v2 (shown here) = curated, smaller, biology-interpretable subset anchored to Approach 1 direction + literature evidence (ECM/Wnt/remodeling + hormone/metabolic candidates).

Not the same as the Kenichi SNS panel: the SNS panel is hypothesis-driven adrenergic/lipolysis sets; this is a menopause-proxy signature.

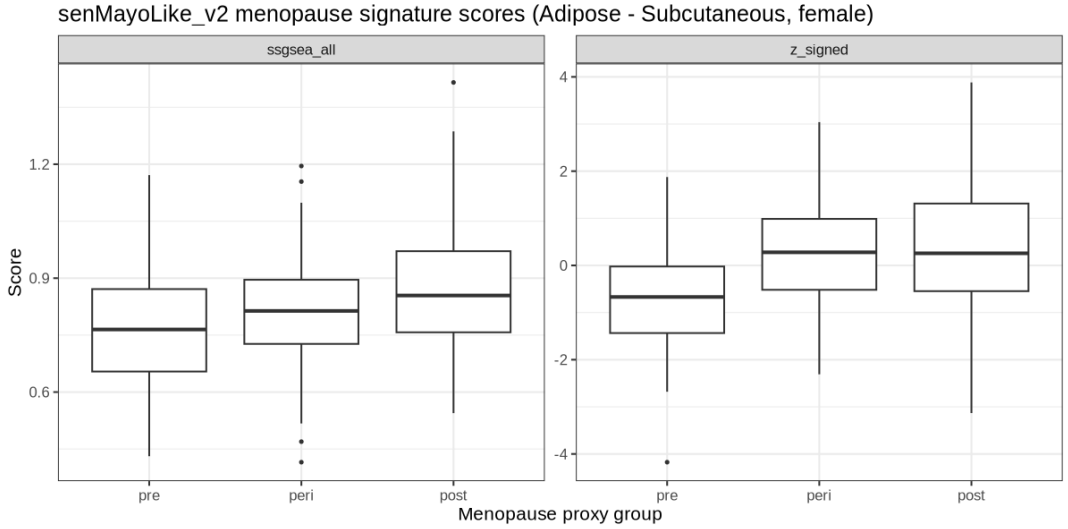
Full per-gene evidence + sources: docs/senMayoLike_v2_signature_details_2026-02-02.xlsx (or CSV).

senMayoLike v2 menopause signature (curated)
Genes and direction from Approach 1

	Direction (in post)	Approach1 log2FC	Approach1 padj
NNAT	down	-1.01	0.004
ACAN	up	1.24	0.020
CEMIP	up	1.42	0.002
COL11A1	up	1.24	0.014
DACT2	up	1.96	0.001
HSD17B2	up	1.26	0.046
MMP3	up	1.42	0.030
PCSK9	up	1.40	0.005
RXFP1	up	1.03	0.033
SCG2	up	1.44	0.001
SFRP2	up	1.51	0.001
STRA6	up	1.64	7.3e-04
XDH	up	1.22	0.029

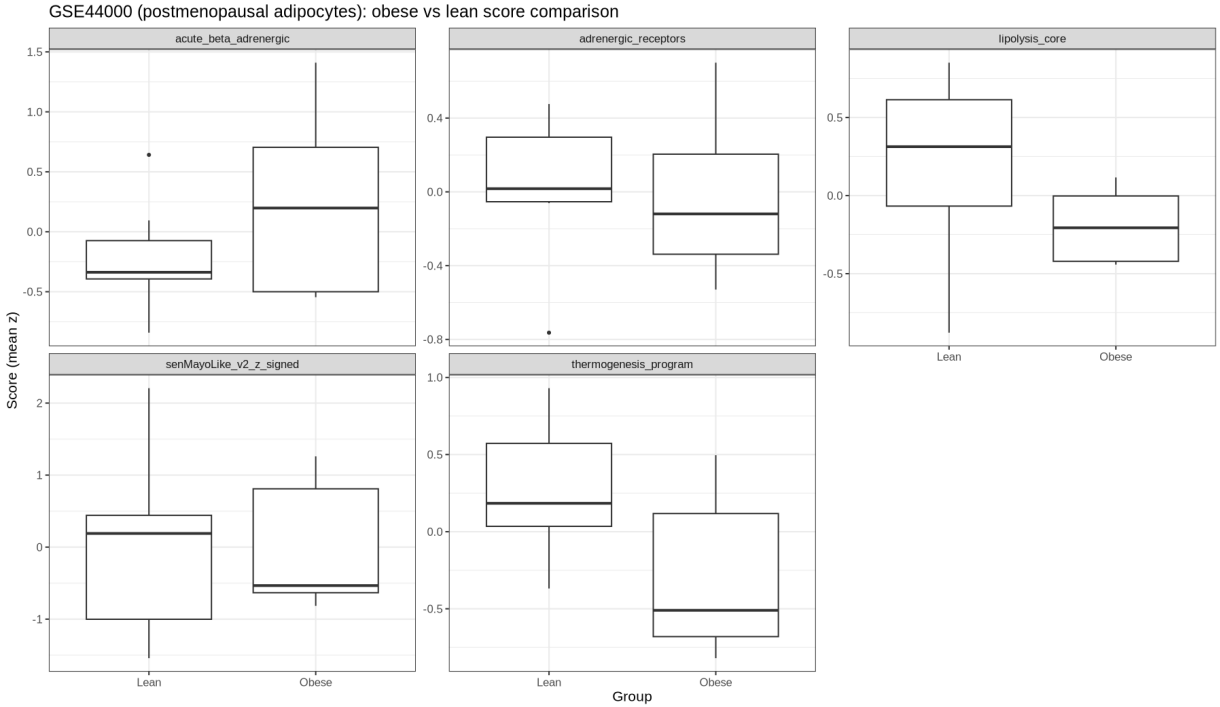
senMayoLike v2 score in GTEx (subcutaneous + visceral)

How the score is computed (Slide 13): - `z_signed` = mean(z of “up” genes) − mean(z of “down” genes), using the same per-gene z-scoring as earlier. - `ssgsea_all` = ssGSEA enrichment score (GSVA::ssgsea) using VST expression. - Conclusion: menopause-proxy groups separate most clearly in subcutaneous (post > pre); visceral is weaker/noisier.



External context (not menopause-status): GSE44000 subcutaneous adipocytes

Dataset: GSE44000 (Agilent microarray)
Tissue/cell type: isolated human subcutaneous adipocytes
Subjects: postmenopausal women only (not a pre/post menopause dataset)
Contrast: obese vs lean (n=14 total)
Purpose here: adipocyte-enriched context to interpret programs without bulk composition confounding



Summary (what we can claim)

- Baseline bulk GTEx adipose does not show robust ‘acute SNS activation’ immediate-early signatures across menopause-proxy groups (NR4A/FOS/JUN/EGR1 set is weak at baseline).
- Across multiple stress-tests, we do observe shifts consistent with altered adrenergic signaling *responsiveness* (cAMP/feedback modules) and depot remodeling, which is compatible with a chronic SNS / catecholamine-resistance framework (but may not directly measure SNS outflow).
- Subcutaneous bulk lipolysis_core differences are strongly composition-sensitive; after adipocyte/SVF proxy adjustment the residual post–pre difference is small → avoid per-adipocyte claims from bulk alone.
- Visceral depot shows weaker/noisier signals overall, but thermogenesis-program shifts are more consistent than lipolysis-core shifts.
- Positive finding: the evidence-backed senMayoLike v2 menopause signature (13 genes) separates menopause-proxy groups (stronger in subcutaneous).
- Recommended wording: ‘Human baseline adipose transcriptomes support menopause-associated remodeling and altered adrenergic/cAMP program behavior; acute SNS activation signatures are not robustly observed in unstimulated tissue.’

Files to share with collaborators

- Master summary: `docs/master_summary_menopause_sns_gtex_2026-02-01.md`
- senMayoLike v2 (full details + evidence):
  - TSV: `references/menopause_signature_senMayoLike_v2_evidence.tsv`
  - CSV: `docs/senMayoLike_v2_signature_details_2026-02-02.csv`
  - Excel: `docs/senMayoLike_v2_signature_details_2026-02-02.xlsx`
- Key figures:
  - `GTEx_v10_AT_analysis_out/figs/kenichi_sns_panel/figure_candidate__lipolysis_thermogenesis.png`
  - `GTEx_v10_AT_analysis_out/figs/next_steps_2026-02-01_composition/kenichi_gene_set_effects_with_composition_adjustment.png`
  - `GTEx_v10_AT_analysis_out/figs/next_steps_2026-02-01_signature_refinement/senMayoLike_v2_scores_by_group.png`